

AI 모델 성능 개선 결과

전체 강우

학습데이터

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)

-
- learning_rate = 0.0001 / 0.0005
 - Loss_function = MSE / MAE
 - Seq_length = 36 (6시간)
 - Num_of_layer = 18

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

학습 , 검증 : 총 1420개 中 1408개 데이터 파일 사용 (결측 5개)

테스트 : 학습 데이터에서 제외한 7개 데이터 파일 (4개이벤트)

+ 학습에서 검증에 사용된 3개 데이터 파일 (2개이벤트) 사용

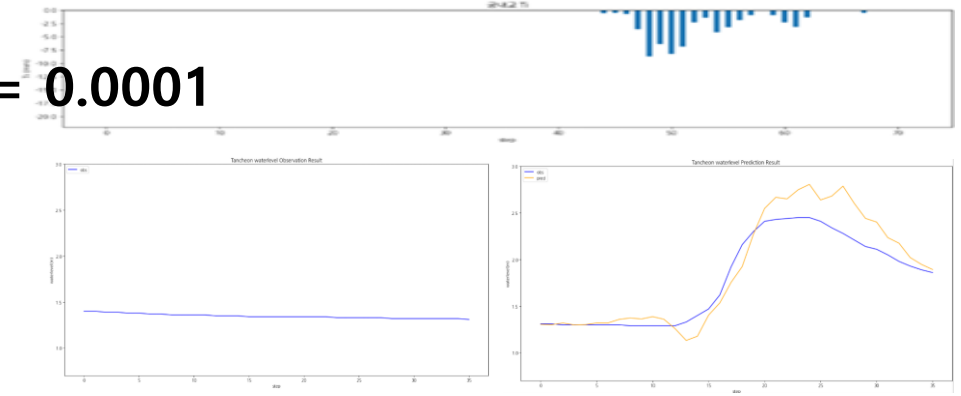
- 학습 : 2014 ~ 2022 67번 (986개)
- 검증 : ~ 2024 97번 (282개)
- 테스트 : ~ 2025 109번 (140개)

num_of_layer = 18, loss_function = MSE, learning_rate = 0.0001

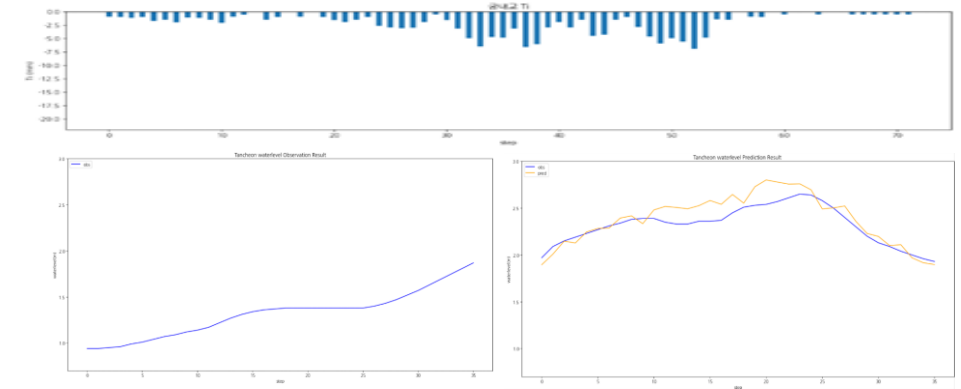
예측 코드 - **궁내교 1**

	궁내교
testdata_200802	r2 : 0.8168 mse : 0.03872 mae : 0.1487 rmse : 0.1967
testdata_220712	r2 : 0.6552 mse : 0.0136 mae : 0.0930 rmse : 0.1169
testdata_230711	r2 : 0.6495 mse : 0.0980 mae : 0.2162 rmse : 0.3131

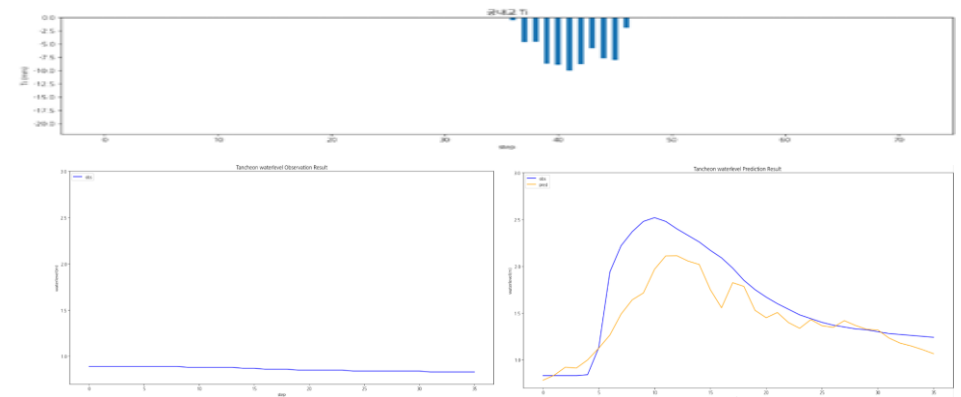
testdata_200802



testdata_220712



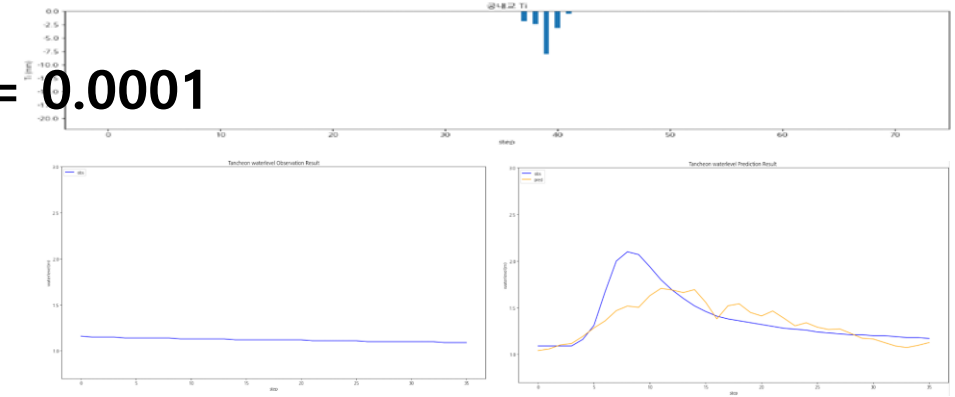
testdata_230711



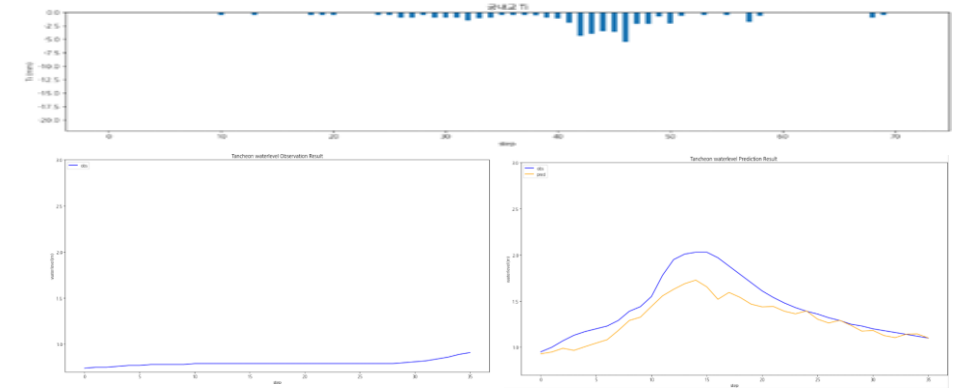
num_of_layer = 18, loss_function = MSE, learning_rate = 0.0001

예측 코드 - **궁내교 2**

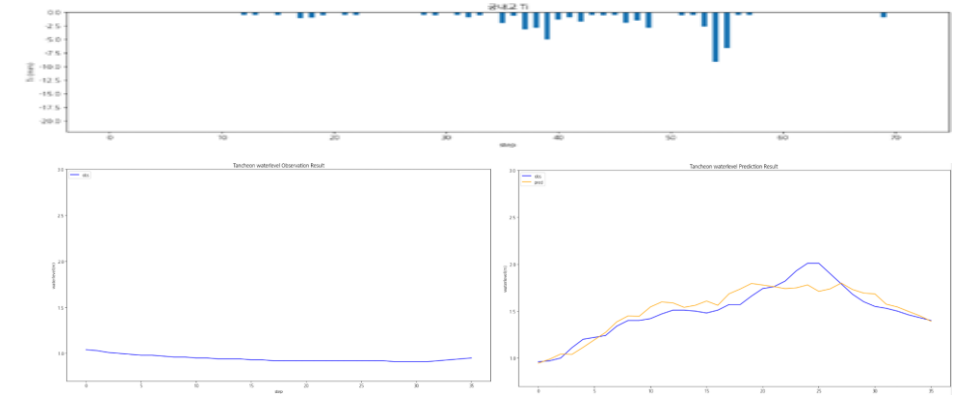
testdata_240723



testdata_250716



testdata_250814



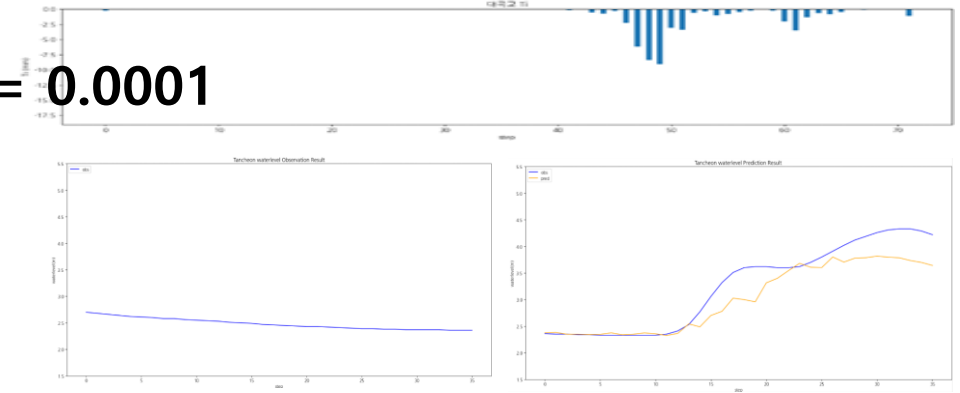
	궁내교
testdata_240723	r2 : 0.5333 mse : 0.0377 mae : 0.1245 rmse : 0.1943
testdata_250716	r2 : 0.6917 mse : 0.0309 mae : 0.1317 rmse : 0.1758
testdata_250814	r2 : 0.8467 mse : 0.0108 mae : 0.0800 rmse : 0.1039

num_of_layer = 18, loss_function = MSE, learning_rate = 0.0001

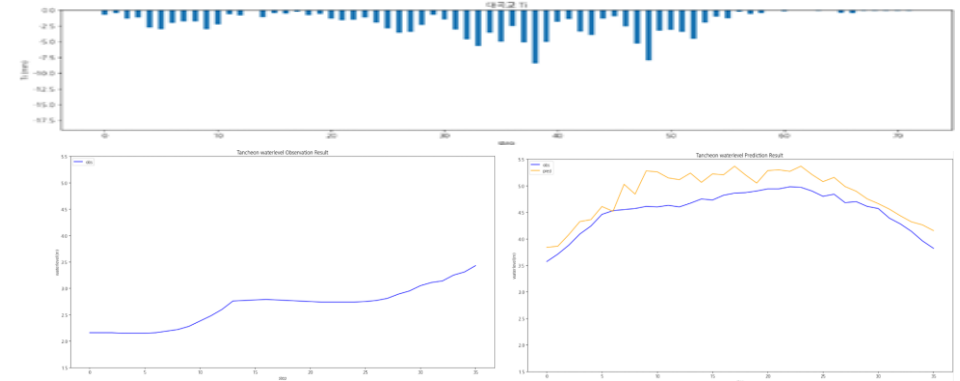
예측 코드 - 대곡교 1

	대곡교
testdata_200802	r2 : 0.8220 mse : 0.1085 mae : 0.2380 rmse : 0.3294
testdata_220712	r2 : 0.1371 mse : 0.1208 mae : 0.3097 rmse : 0.3476
testdata_230711	r2 : 0.6871 mse : 0.2847 mae : 0.4375 rmse : 0.5335

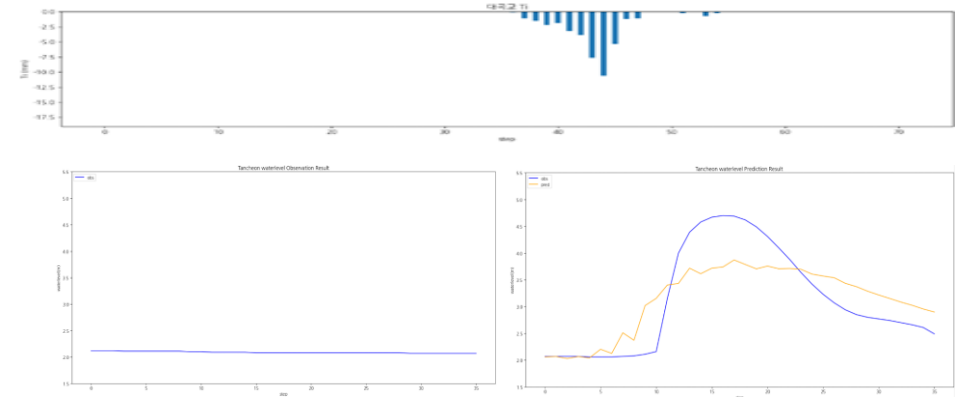
testdata_200802



testdata_220712



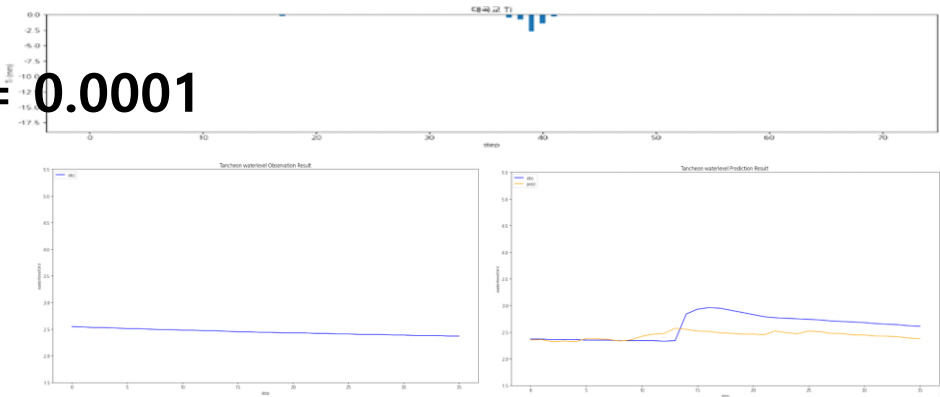
testdata_230711



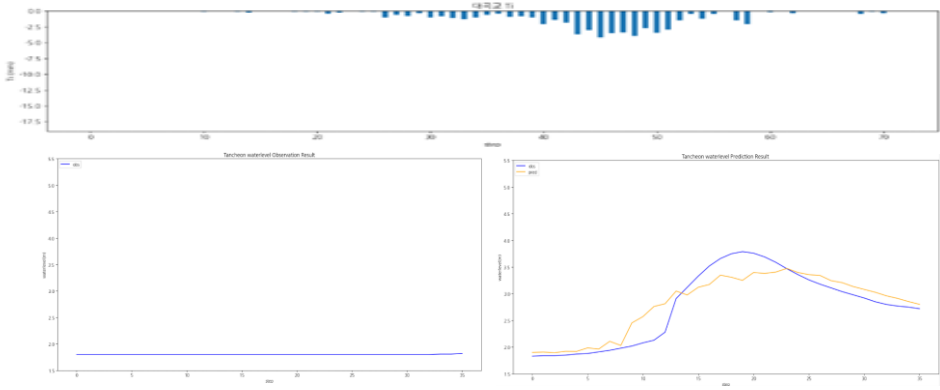
num_of_layer = 18, loss_function = MSE, learning_rate = 0.0001
예측 코드 - 대곡교 2

	대곡교
testdata_240723	r2 : -0.2634 mse : 0.0600 mae : 0.2021 rmse : 0.2450
testdata_250716	r2 : 0.8524 mse : 0.0681 mae : 0.2043 rmse : 0.2610
testdata_250814	r2 : 0.7143 mse : 0.0991 mae : 0.2523 rmse : 0.3148

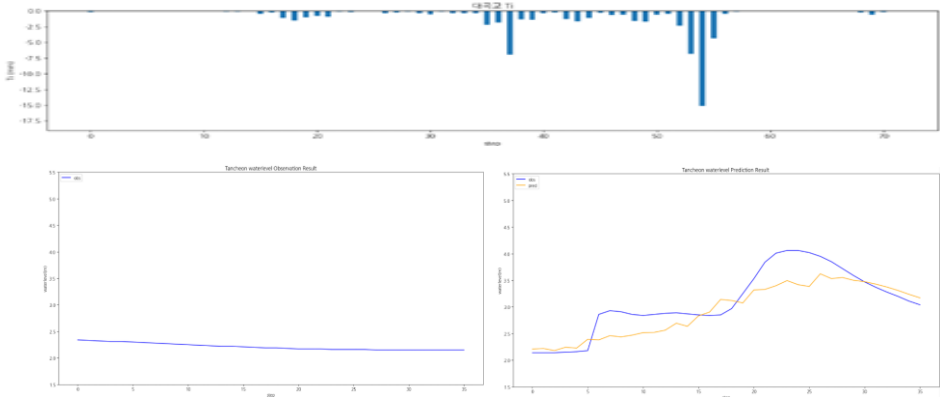
testdata_240723



testdata_250716



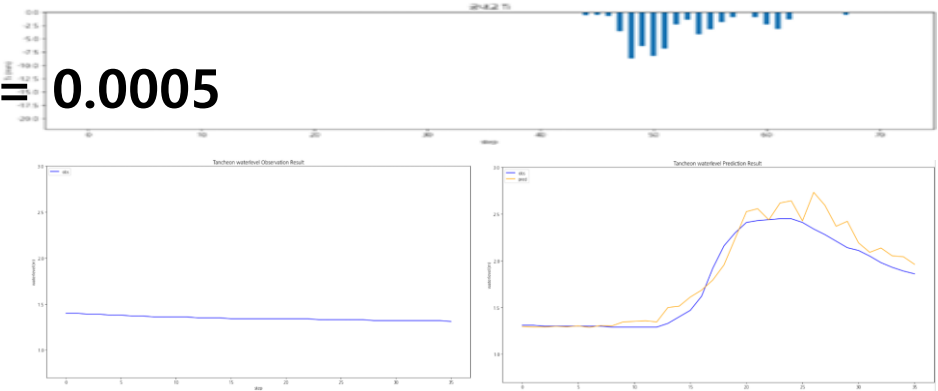
testdata_250814



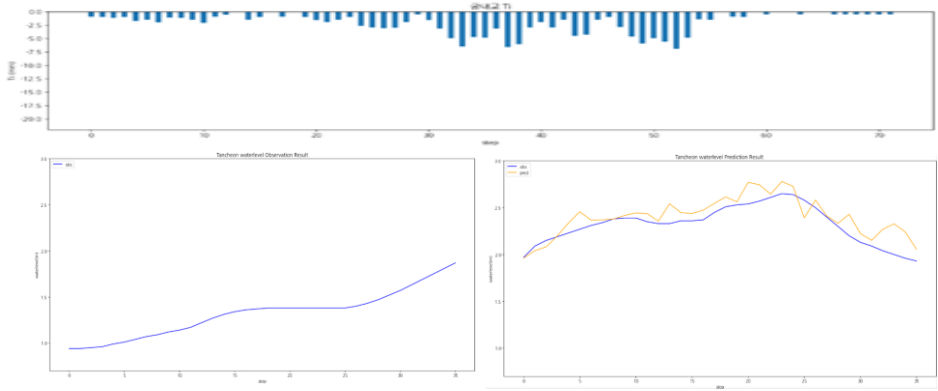
num_of_layer = 18 , loss_function = MAE, learning_rate = 0.0005
예측 코드 - **궁내교 1**

	궁내교
testdata_200802	r2 : 0.9110 mse : 0.0188 mae : 0.1008 rmse : 0.1372
testdata_220712	r2 : 0.5566 mse : 0.0175 mae : 0.1040 rmse : 0.1040
testdata_230711	r2 : 0.7232 mse : 0.0774 mae : 0.2122 rmse : 0.2782

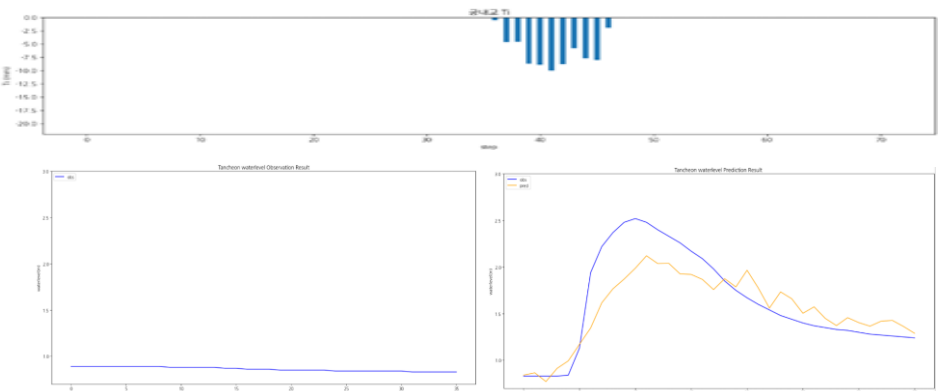
testdata_200802



testdata_220712



testdata_230711

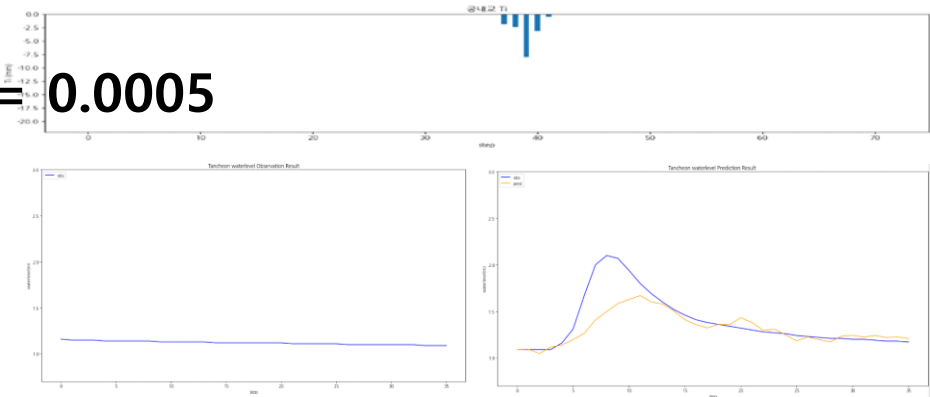


* model_(gn/dg)_all_mae

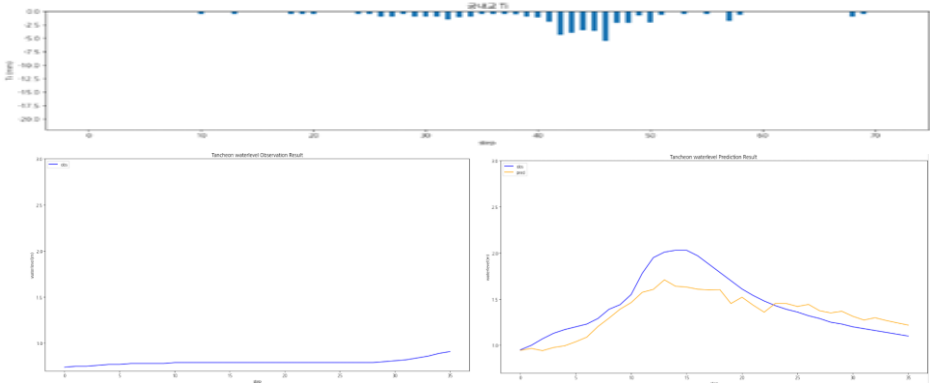
num_of_layer = 18 , loss_function = MAE, learning_rate = 0.0005
예측 코드 - 궁내교 2

	궁내교
testdata_240723	r2 : 0.5535 mse : 0.0361 mae : 0.1023 rmse : 0.1900
testdata_250716	r2 : 0.6694 mse : 0.0331 mae : 0.1517 rmse : 0.1821
testdata_250814	r2 : 0.6832 mse : 0.0223 mae : 0.1198 rmse : 0.1494

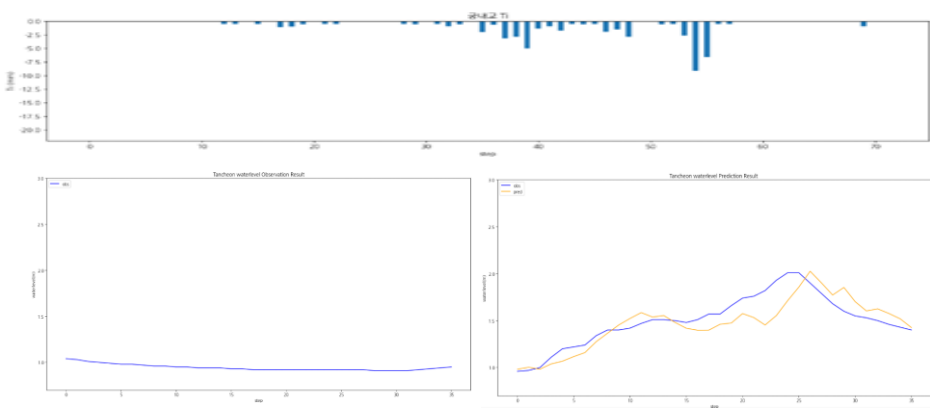
testdata_240723



testdata_250716



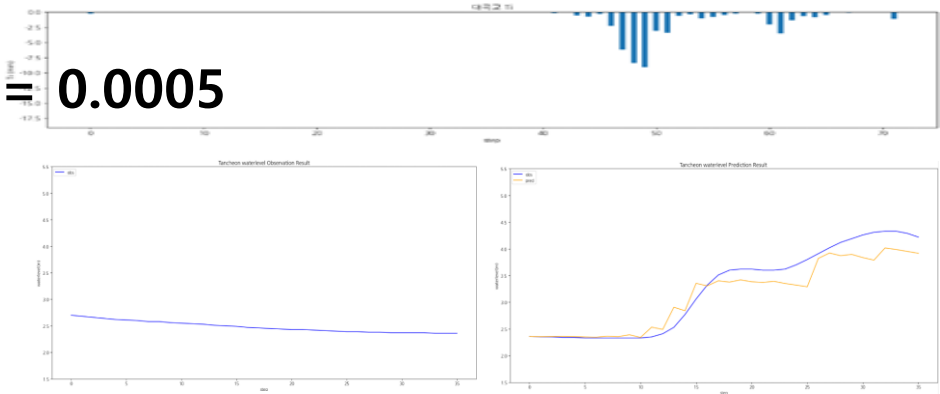
testdata_250814



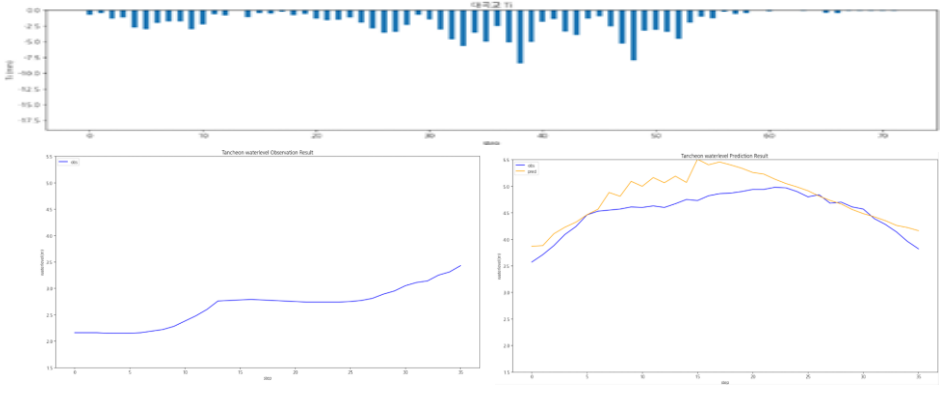
num_of_layer = 18 , loss_function = MAE, learning_rate = 0.0005
예측 코드 - 대곡교 1

	대곡교
testdata_200802	r2 : 0.9067 mse : 0.0569 mae : 0.1824 rmse : 0.2385
testdata_220712	r2 : 0.2335 mse : 0.1073 mae : 0.2582 rmse : 0.3276
testdata_230711	r2 : 0.6120 mse : 0.3530 mae : 0.4811 rmse : 0.5942

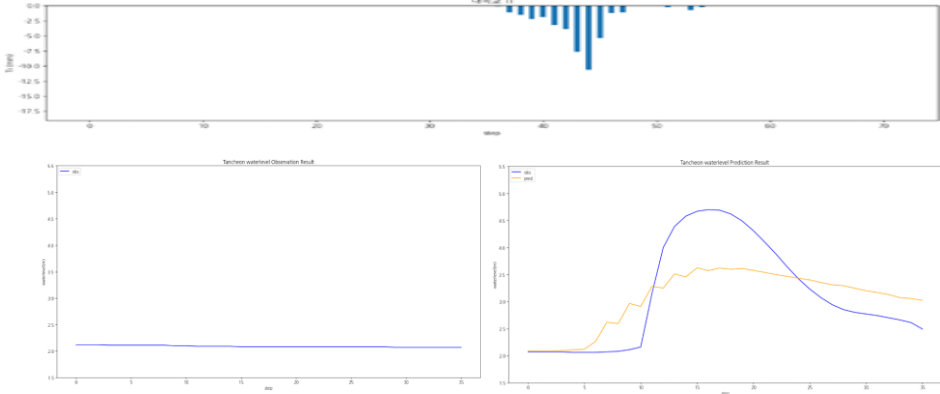
testdata_200802



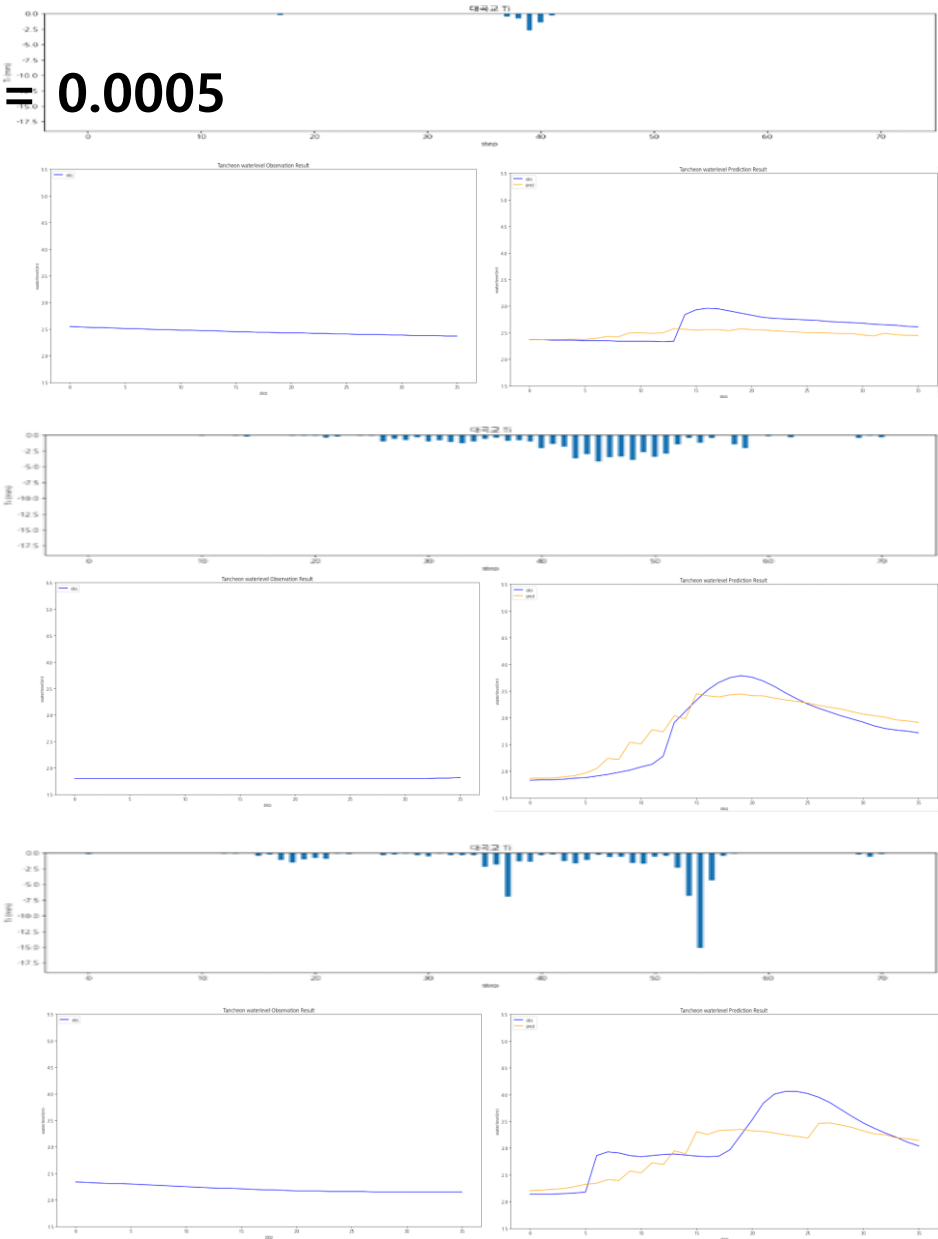
testdata_220712



testdata_230711



num_of_layer = 18 , loss_function = MAE, learning_rate = 0.0005
예측 코드 - 대곡교 2



	대곡교
testdata_240723	r2 : 0.0143 mse : 0.0468 mae : 0.1858 rmse : 0.2164
testdata_250716	r2 : 0.8707 mse : 0.0596 mae : 0.1949 rmse : 0.2443
testdata_250814	r2 : 0.5747 mse : 0.1475 mae : 0.2976 rmse : 0.3841

testdata_240723

testdata_250716

testdata_250814

누적 강우 110mm 이상 4 , 5

학습데이터

같은 연도.다른 강우 패턴 대응 가능성 검증

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- 그래프 스케일 고정 : 궁내교 (0.7 - 3) 대곡교 (1.5 - 5.5)

-
- learning_rate = 0.0001
 - Loss_function = MSE
 - Seq_length = 36 (6시간)
 - Num_of_layer = 18

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출 → 파일 단위 누적 강우 110mm 이상 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

학습 , 검증 : 총 11개 中 9개 데이터 파일 사용

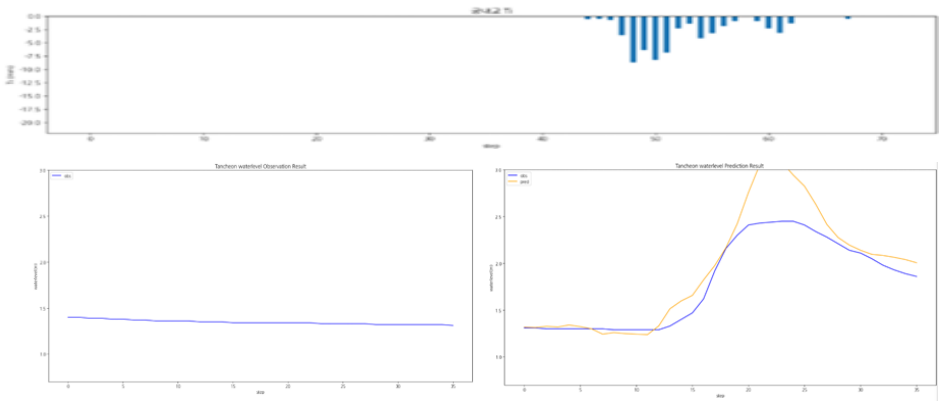
테스트 : 학습 데이터에서 제외한 2개 데이터 파일 + 4개 이벤트 사용

- 학습 : 2014 ~ 2024 (8개)
- 검증 : 2022(39번이벤트) (1개)
- 테스트 : 2022(66.67번이벤트) (1개)

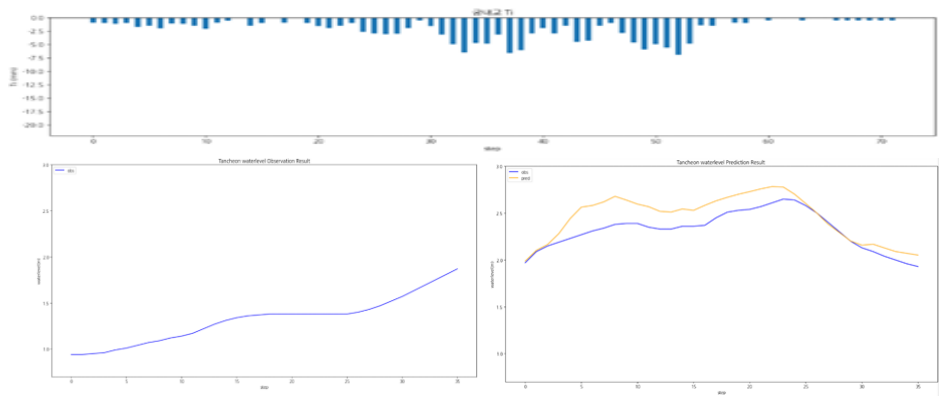
누적 강우 110mm 이상 4 예측 코드 - **궁내교 1**

	궁내교
testdata_200802	r2 : 0.7237 mse : 0.0584 mae : 0.1572 rmse : 0.2417
testdata_220712	r2 : 0.3192 mse : 0.0270 mae : 0.1369 rmse : 0.1643
testdata_230711	r2 : 0.8533 mse : 0.0410 mae : 0.1509 rmse : 0.2025

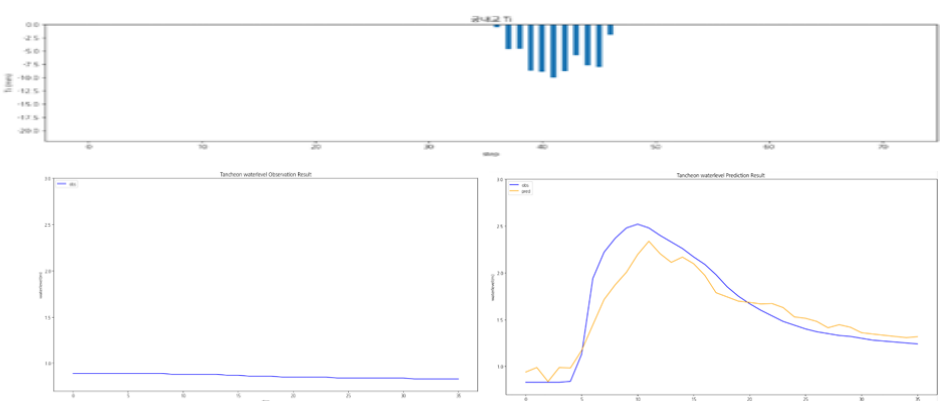
testdata_200802



testdata_220712



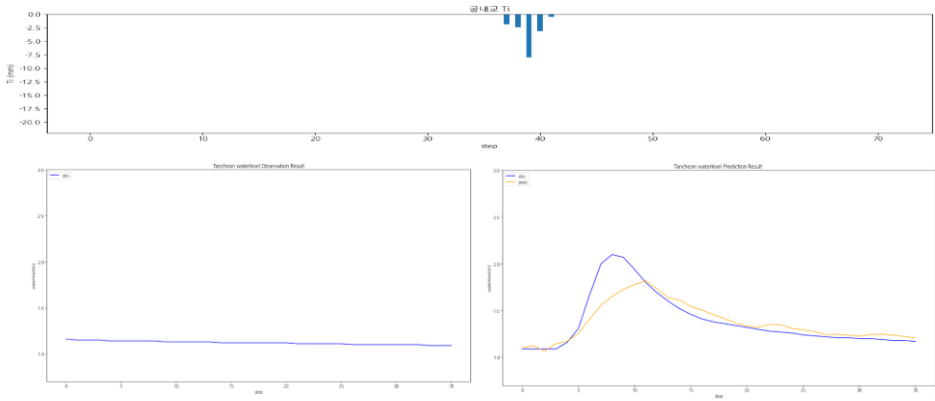
testdata_230711



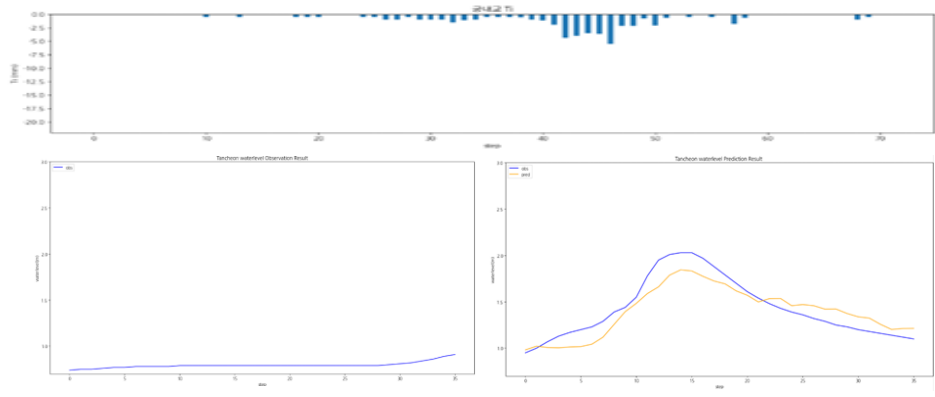
누적 강우 110mm 이상 4 예측 코드 - **궁내교 2**

	궁내교
testdata_240723	r2 : 0.7623 mse : 0.0192 mae : 0.0851 rmse : 0.1386
testdata_250716	r2 : 0.8108 mse : 0.0189 mae : 0.1236 rmse : 0.1377
testdata_250814	r2 : 0.8475 mse : 0.0107 mae : 0.0815 rmse : 0.1036

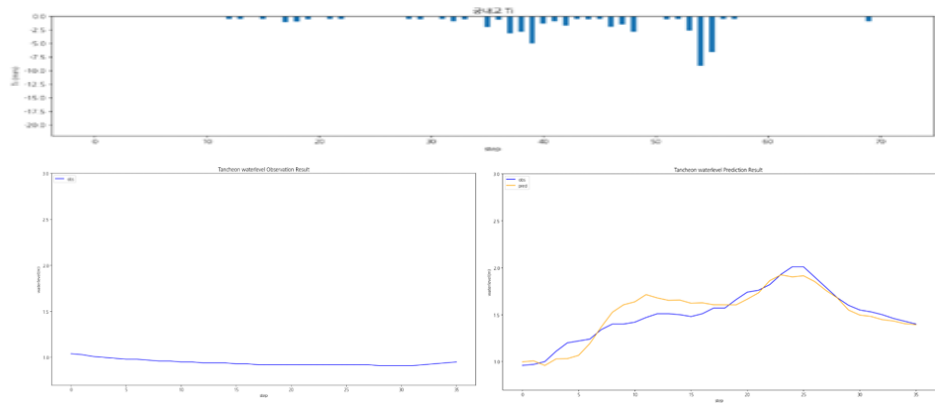
testdata_240723



testdata_250716



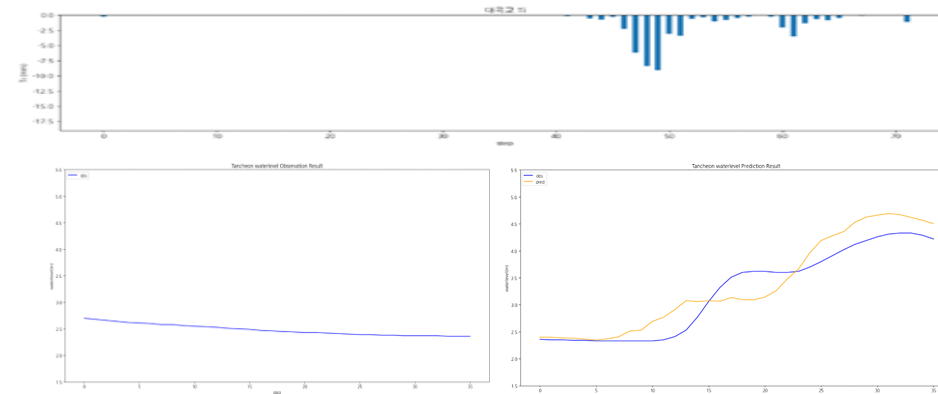
testdata_250814



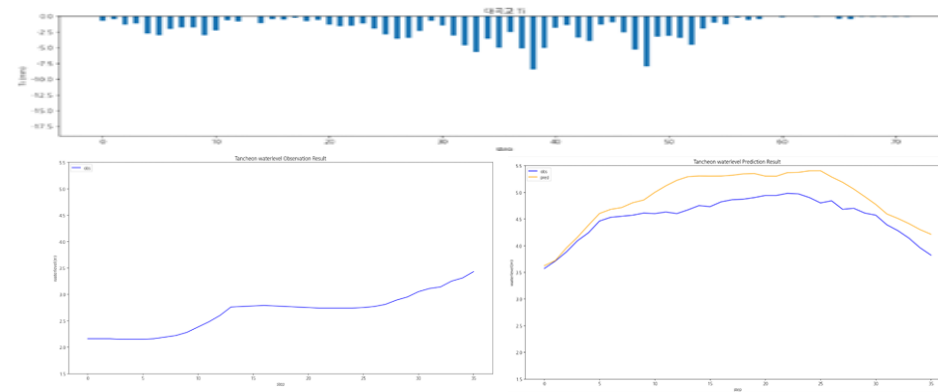
누적 강우 110mm 이상 4 예측 코드 - 대곡교 1

	대곡교
testdata_200802	r2 : 0.8350 mse : 0.1006 mae : 0.2683 rmse : 0.3172
testdata_220712	r2 : -0.0424 mse : 0.1459 mae : 0.3414 rmse : 0.3820
testdata_230711	r2 : 0.7456 mse : 0.2314 mae : 0.3847 rmse : 0.4811

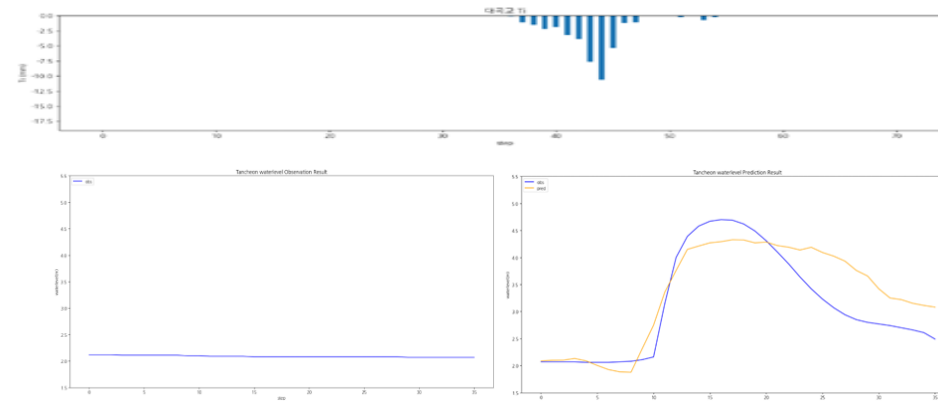
testdata_200802



testdata_220712



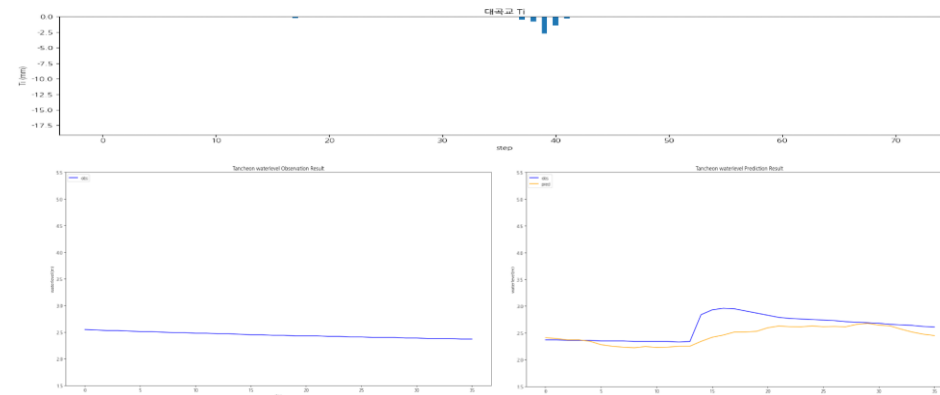
testdata_230711



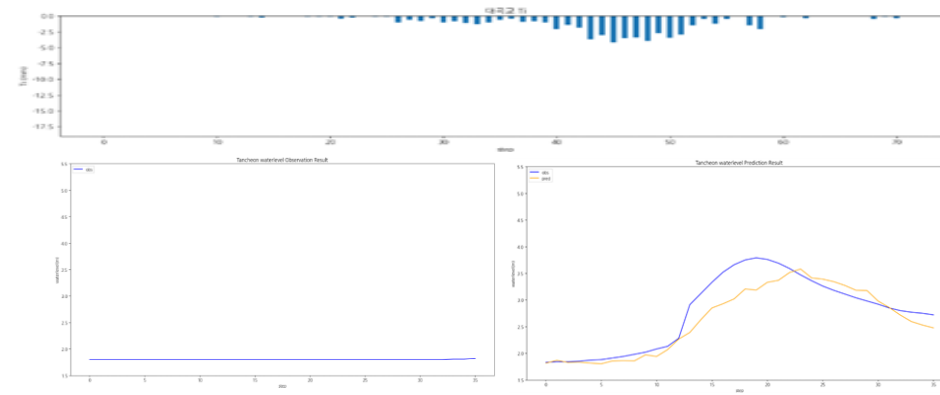
누적 강우 110mm 이상 4 예측 코드 - 대곡교 2

	대곡교
testdata_240723	r2 : 0.0871 mse : 0.0434 mae : 0.1510 rmse : 0.2083
testdata_250716	r2 : 0.8305 mse : 0.0782 mae : 0.1994 rmse : 0.2797
testdata_250814	r2 : 0.5011 mse : 0.1731 mae : 0.3295 rmse : 0.4160

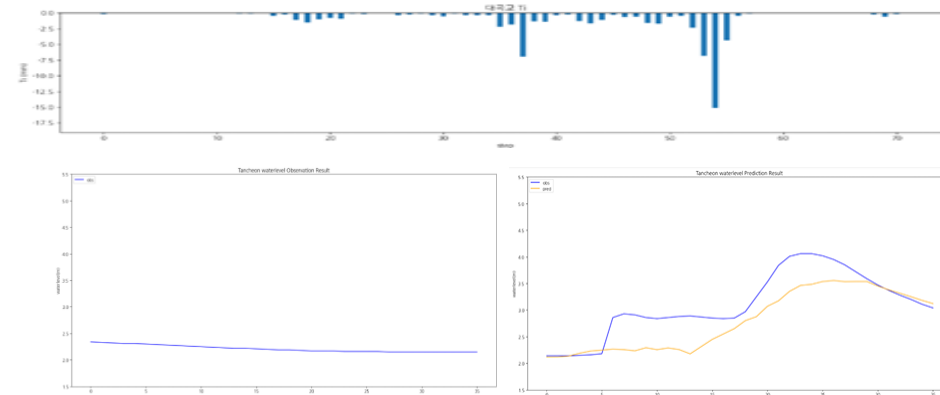
testdata_240723



testdata_250716



testdata_250814



학습데이터

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- 그래프 스케일 고정 : 궁내교 (0.7 - 3) 대곡교 (1.5 - 5.5)

-
- learning_rate = 0.0001
 - Loss_function = MSE
 - Seq_length = 36 (6시간)
 - Num_of_layer = 18

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출 → 파일 단위 누적 강우 110mm 이상 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

학습 , 검증 : 총 11개 中 9개 데이터 파일 사용

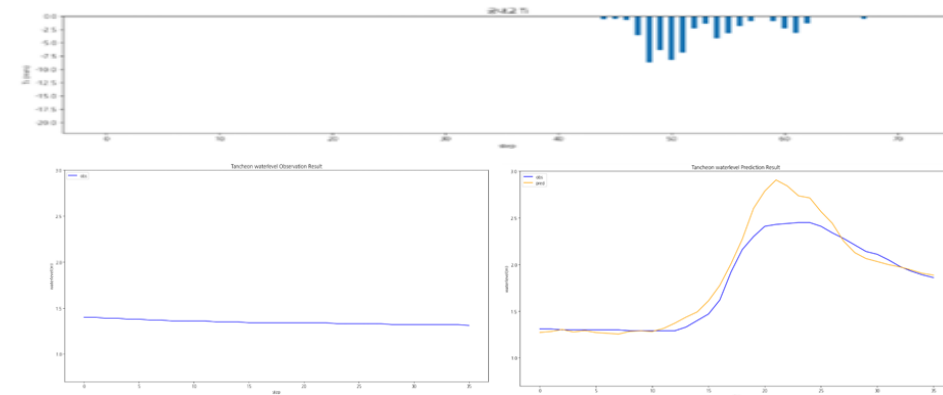
테스트 : 학습 데이터에서 제외한 2개 데이터 파일 + 4개 이벤트 사용

- 학습 : 2014 ~ 2024 (8개)
- 검증 : 2022(35번이벤트) (1개)
- 테스트 : 2022(66.67번이벤트) (1개)

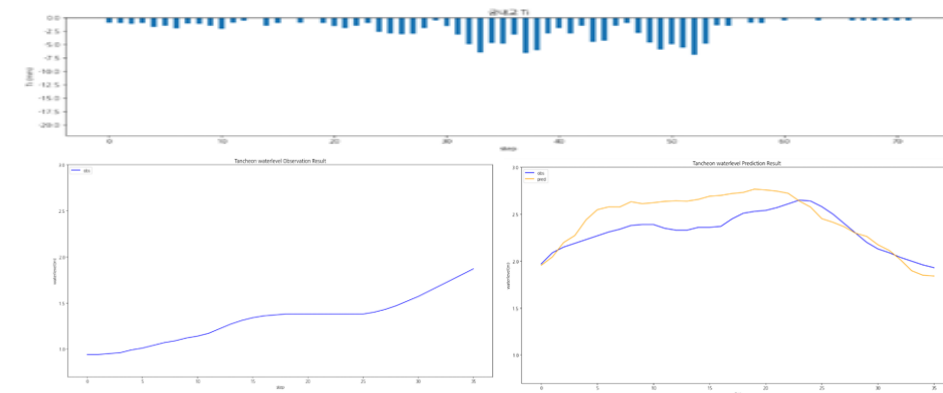
누적 강우 110mm 이상 5 예측 코드 - **궁내교 1**

	궁내교
testdata_200802	r2 : 0.8774 mse : 0.0259 mae : 0.1047 rmse : 0.1609
testdata_220712	r2 : 0.0586 mse : 0.0373 mae : 0.1603 rmse : 0.1932
testdata_230711	r2 : 0.8028 mse : 0.0851 mae : 0.1455 rmse : 0.2349

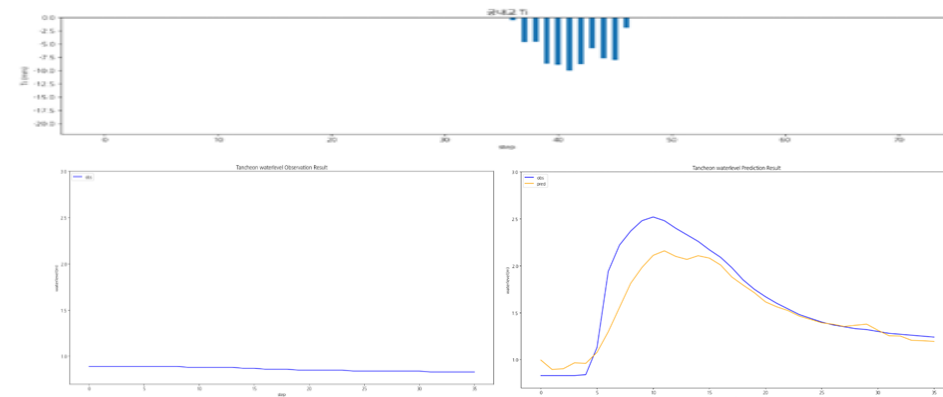
testdata_200802



testdata_220712



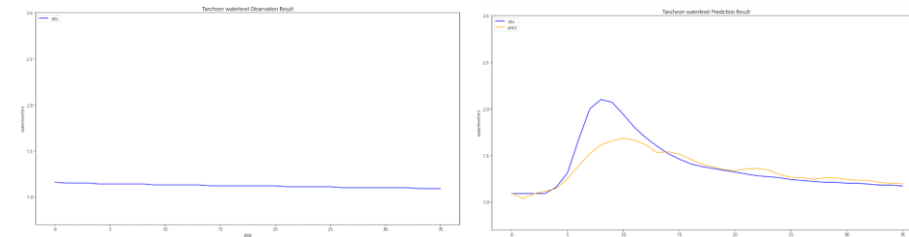
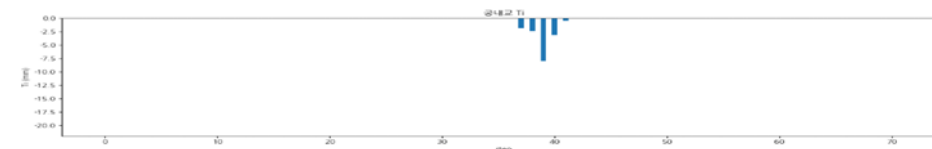
testdata_230711



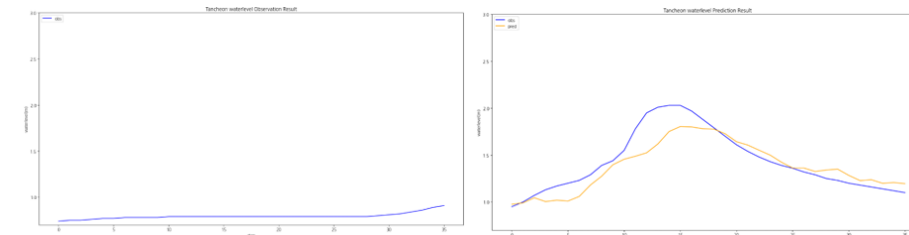
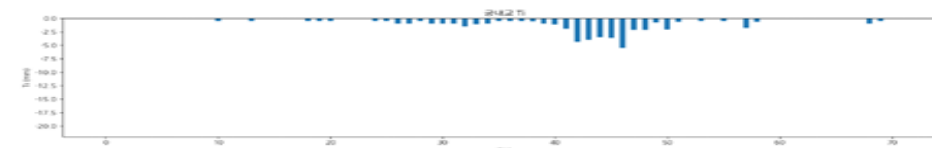
누적 강우 110mm 이상 5 예측 코드 - **궁내교 2**

	궁내교
testdata_240723	r2 : 0.7051 mse : 0.0238 mae : 0.0874 rmse : 0.1544
testdata_250716	r2 : 0.7764 mse : 0.0224 mae : 0.1110 rmse : 0.1497
testdata_250814	r2 : 0.8673 mse : 0.0093 mae : 0.0756 rmse : 0.0967

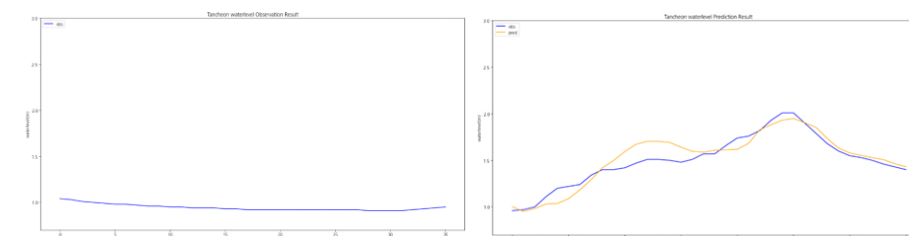
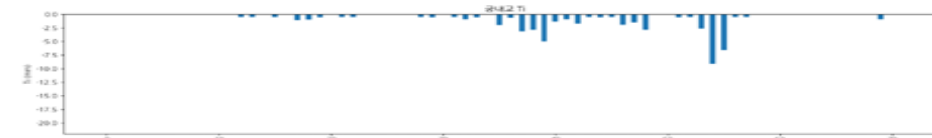
testdata_240723



testdata_250716



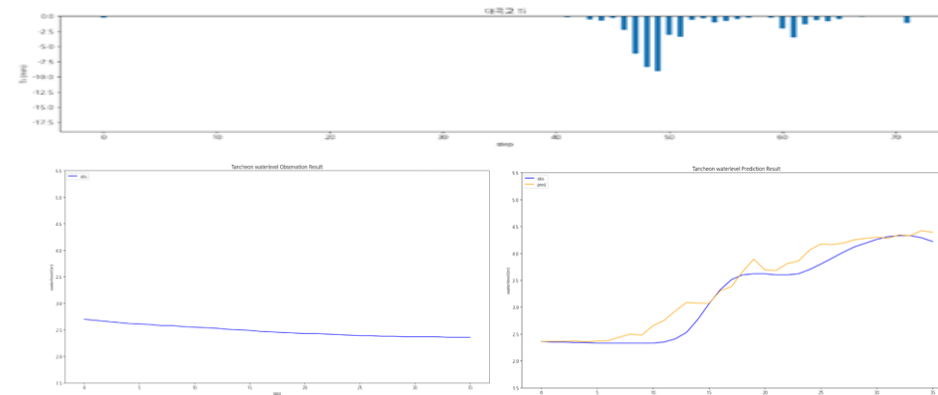
testdata_250814



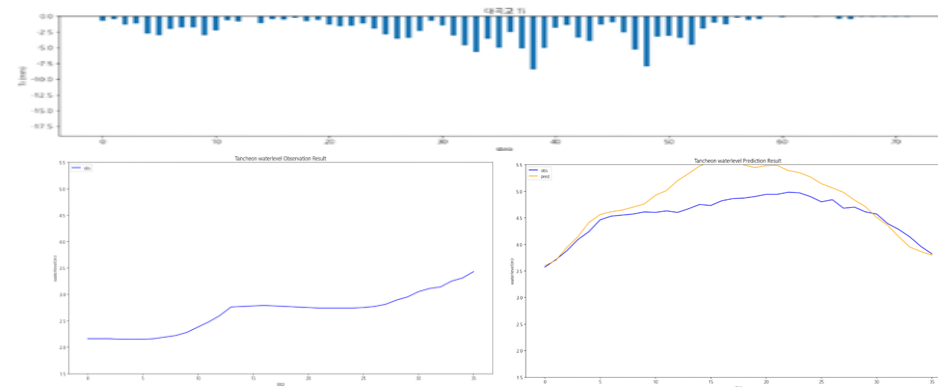
누적 강우 110mm 이상 5 예측 코드 - 대곡교 1

	대곡교
testdata_200802	r2 : 0.9246 mse : 0.0460 mae : 0.1541 rmse : 0.2144
testdata_220712	r2 : -0.1000 mse : 0.1540 mae : 0.3041 rmse : 0.3924
testdata_230711	r2 : 0.8276 mse : 0.1568 mae : 0.3195 rmse : 0.3961

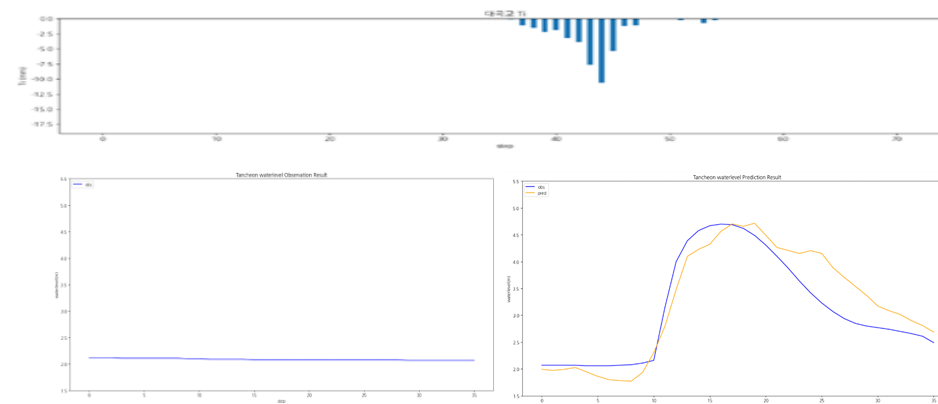
testdata_200802



testdata_220712



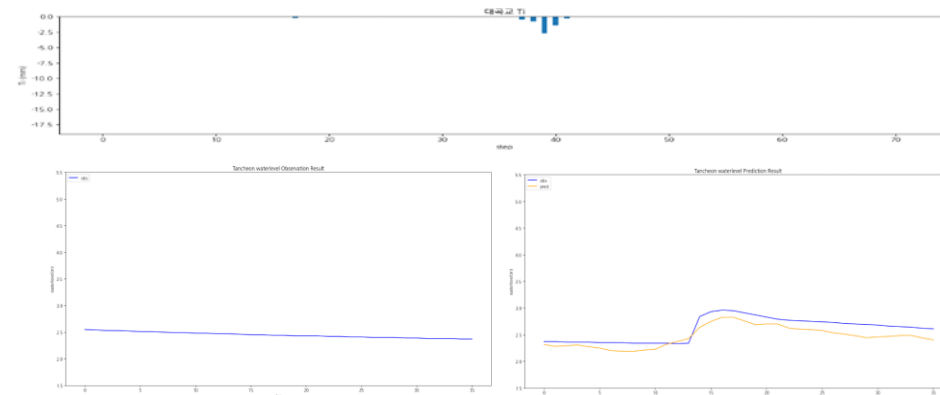
testdata_230711



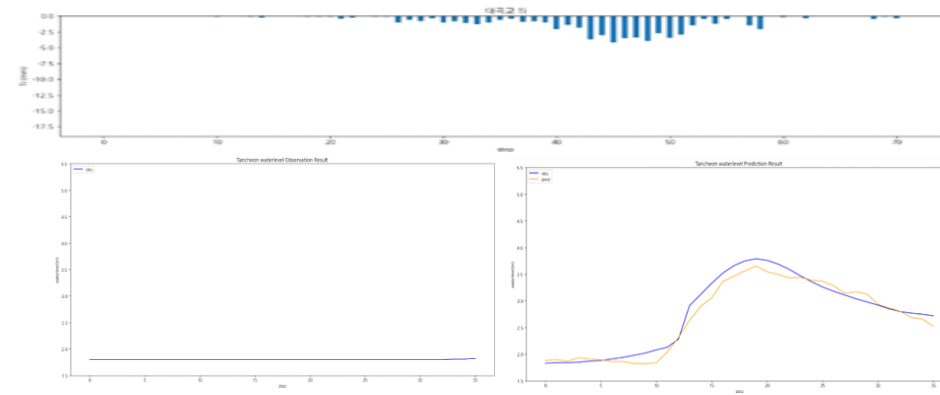
누적 강우 110mm 이상 5 예측 코드 - 대곡교 2

	대곡교
testdata_240723	r2 : 0.5130 mse : 0.0231 mae : 0.1419 rmse : 0.1521
testdata_250716	r2 : 0.9577 mse : 0.0195 mae : 0.1145 rmse : 0.1397
testdata_250814	r2 : 0.6115 mse : 0.1348 mae : 0.3007 rmse : 0.3671

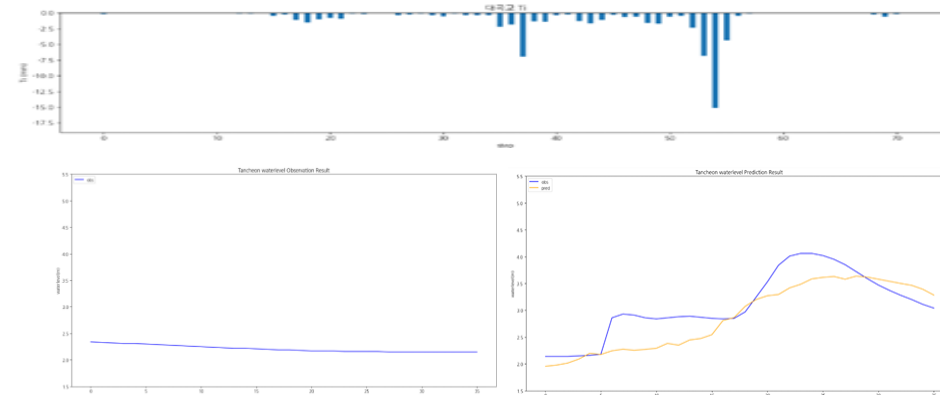
testdata_240723



testdata_250716



testdata_250814



관심 수위 이상

학습데이터

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- 그래프 스케일 고정 : 궁내교 (0.7 - 3) 대곡교 (1.5 - 5.5)

-
- learning_rate = 0.0005
 - Loss_function = MAE
 - Seq_length = 36 (6시간)
 - Num_of_layer = 18

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출 → 관심 수위 이상 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

학습 , 검증 : 총 67개 中 61개 데이터 파일 사용

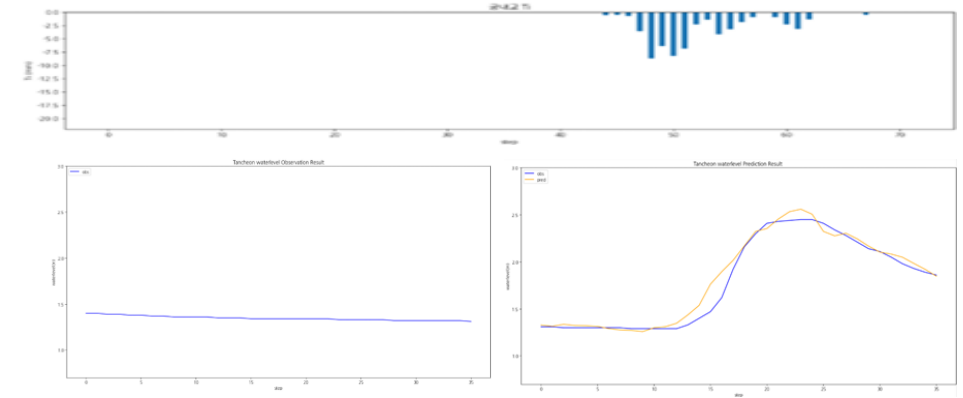
테스트 : 학습 데이터에서 제외한 6개 데이터 파일 (4개 이벤트)

+ 학습에서 검증에 사용된 2개 이벤트 사용

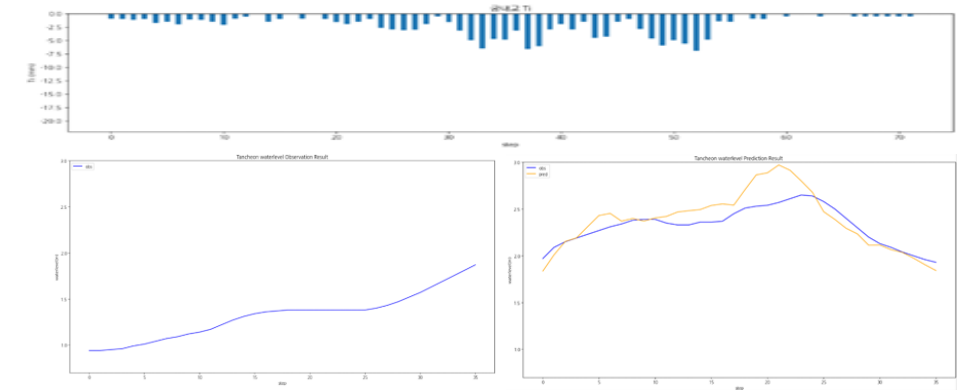
- 학습 : 2014 ~ 20230629 (43개)
- 검증 : 20230713 ~ 250813 (12개)
- 테스트 : 2550916 ~ 251006 (6개)

관심 수위 이상 + testdata 증강 + MAE + 0.0005 예측 코드 - **궁내교 1**

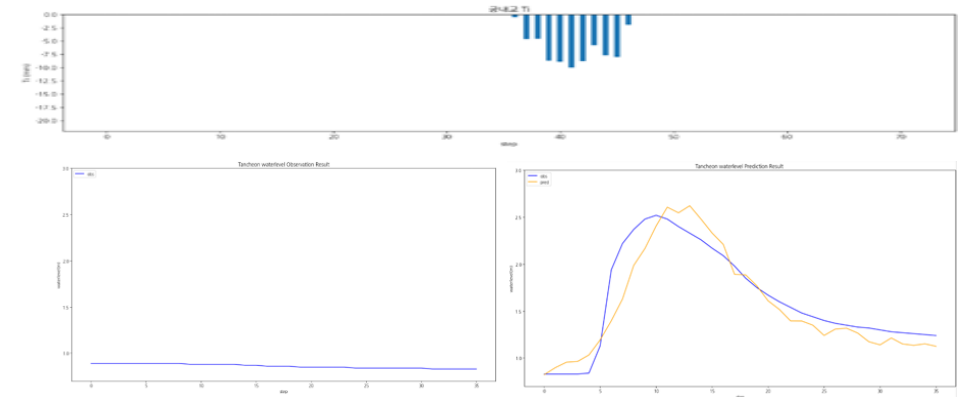
testdata_200802



testdata_220712



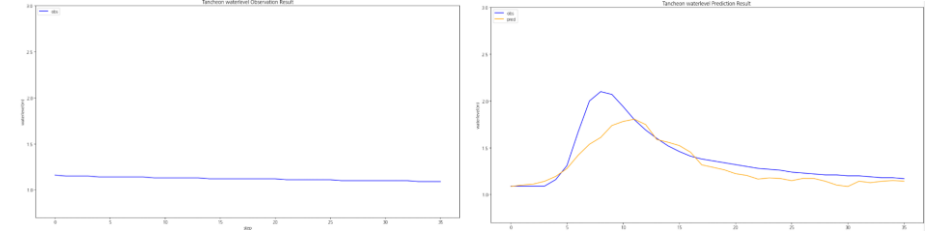
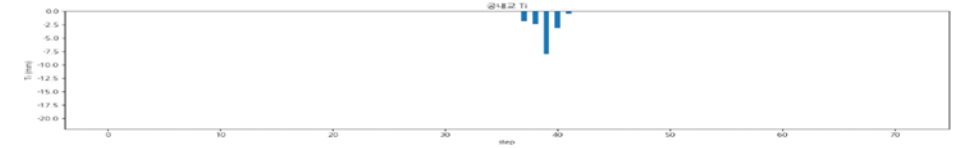
testdata_230711



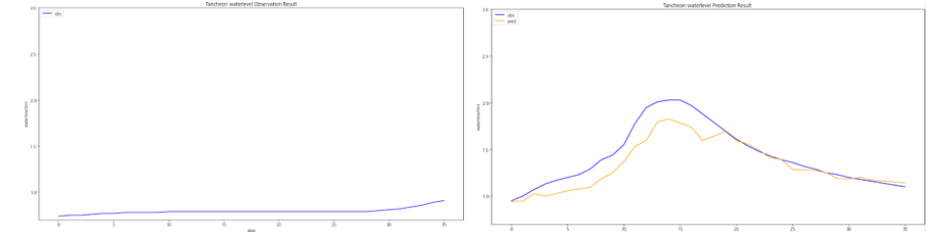
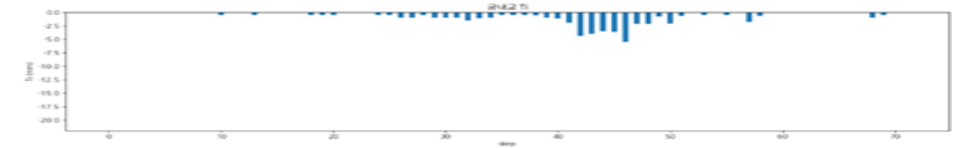
	궁내교
testdata_200802	r2 : 0.9656 mse : 0.0072 mae : 0.0558 rmse : 0.0852
testdata_220712	r2 : 0.4201 mse : 0.0230 mae : 0.1139 rmse : 0.1516
testdata_230711	r2 : 0.8622 mse : 0.0385 mae : 0.1486 rmse : 0.1963

관심 수위 이상 + testdata 증강 + MAE + 0.0005
 예측 코드 - **궁내교 2**

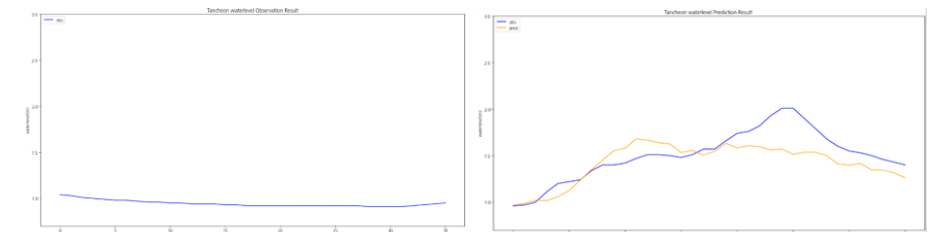
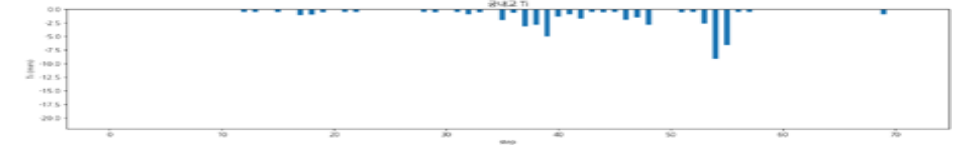
testdata_240723



testdata_250716



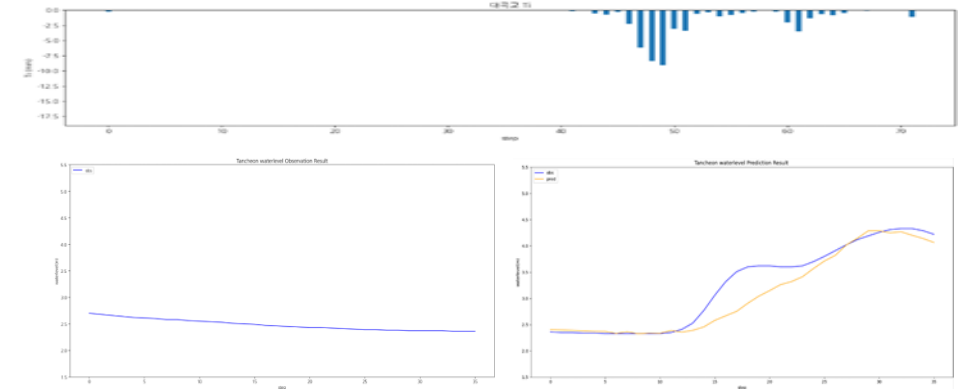
testdata_250814



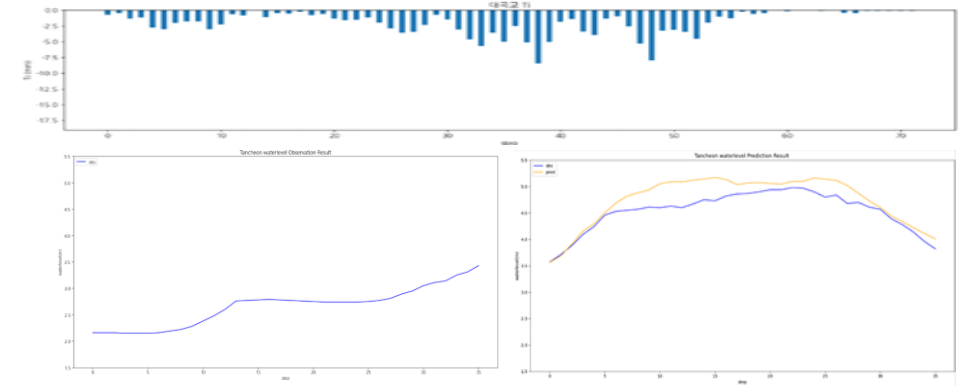
	궁내교
testdata_240723	r2 : 0.7308 mse : 0.0217 mae : 0.0962 rmse : 0.1475
testdata_250716	r2 : 0.7947 mse : 0.0205 mae : 0.1047 rmse : 0.1435
testdata_250814	r2 : 0.5087 mse : 0.0346 mae : 0.1446 rmse : 0.1861

관심 수위 이상 + testdata 증강 + MAE + 0.0005 예측 코드 - 대곡교 1

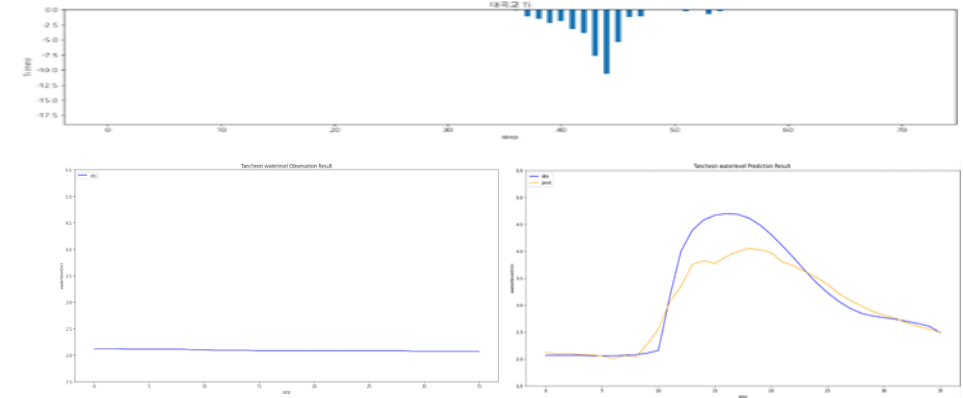
testdata_200802



testdata_220712



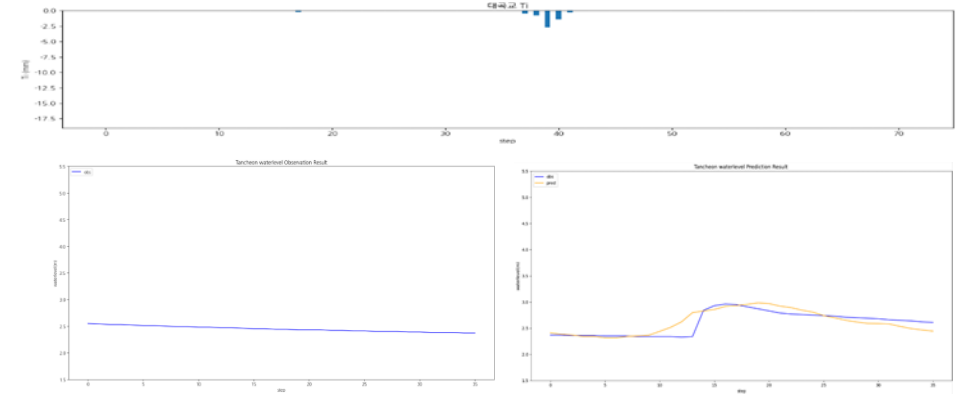
testdata_230711



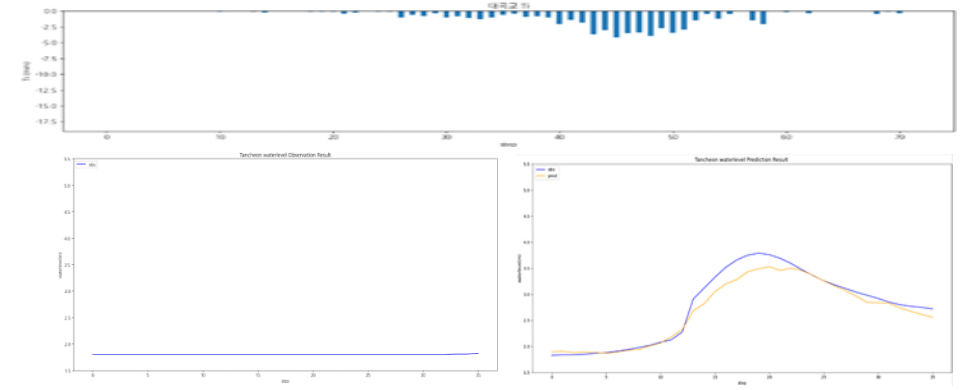
	대곡교
testdata_200802	r2 : 0.8748 mse : 0.0763 mae : 0.1754 rmse : 0.2763
testdata_220712	r2 : 0.5503 mse : 0.0629 mae : 0.2047 rmse : 0.2509
testdata_230711	r2 : 0.8659 mse : 0.1219 mae : 0.2258 rmse : 0.3492

관심 수위 이상 + testdata 증강 + MAE + 0.0005 예측 코드 - 대곡교 2

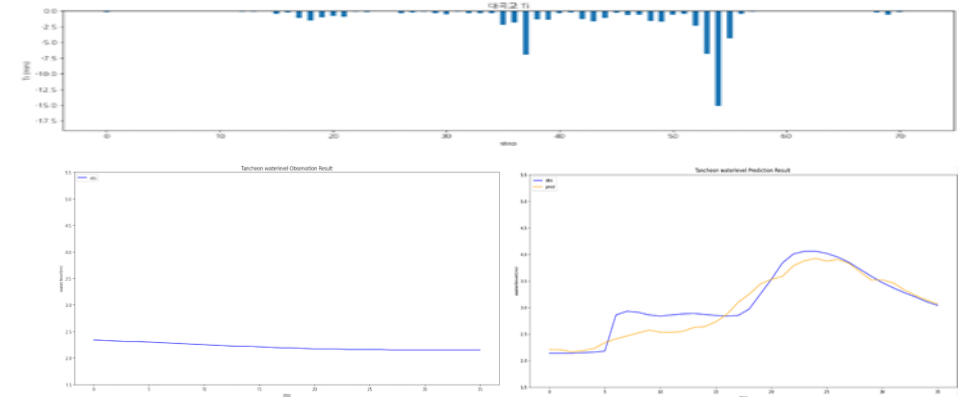
testdata_240723



testdata_250716



testdata_250814



	대곡교
testdata_240723	r2 : 0.6749 mse : 0.0154 mae : 0.0860 rmse : 0.1244
testdata_250716	r2 : 0.9470 mse : 0.0244 mae : 0.1098 rmse : 0.1563
testdata_250814	r2 : 0.8781 mse : 0.0423 mae : 0.1589 rmse : 0.2056

학습데이터

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- 그래프 스케일 고정 : 궁내교 (0.7 - 3) 대곡교 (1.5 - 5.5)

-
- learning_rate = 0.0001
 - Loss_function = Huber
 - Seq_length = 36 (6시간)
 - Num_of_layer = 18

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출 → 관심 수위 이상 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

학습 , 검증 : 총 67개 中 61개 데이터 파일 사용

테스트 : 학습 데이터에서 제외한 6개 데이터 파일 (4개 이벤트)

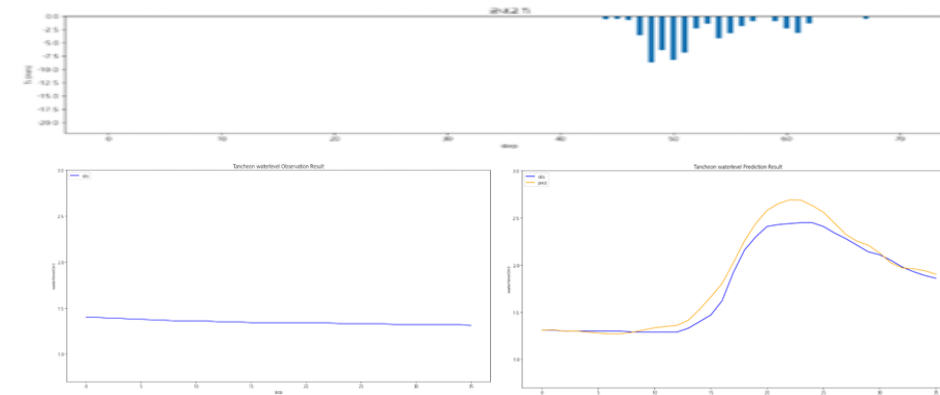
+ 학습에서 검증에 사용된 2개 이벤트 사용

- 학습 : 2014 ~ 20230629 (43개)
- 검증 : 20230713 ~ 250813 (12개)
- 테스트 : 2550916 ~ 251006 (6개)

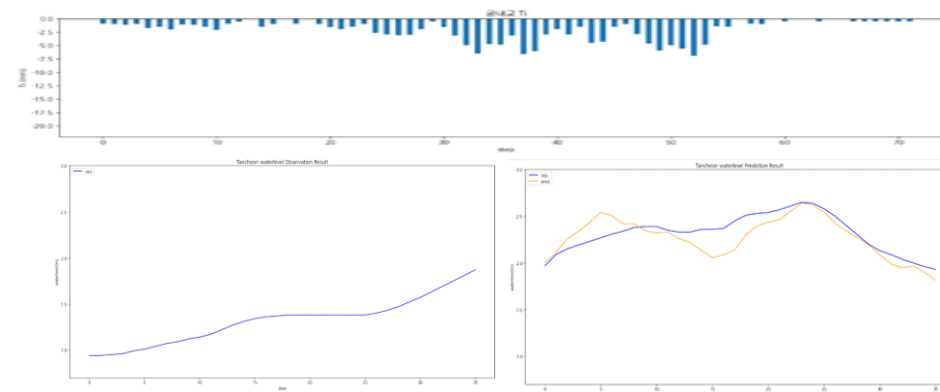
관심 수위 이상 + Huber + 0.0001 예측 코드 - **궁내교 1**

	궁내교
testdata_200802	r2 : 0.9436 mse : 0.0119 mae : 0.0800 rmse : 0.1091
testdata_220712	r2 : 0.5280 mse : 0.0187 mae : 0.1057 rmse : 0.1368
testdata_230711	r2 : 0.8739 mse : 0.0352 mae : 0.1445 rmse : 0.1878

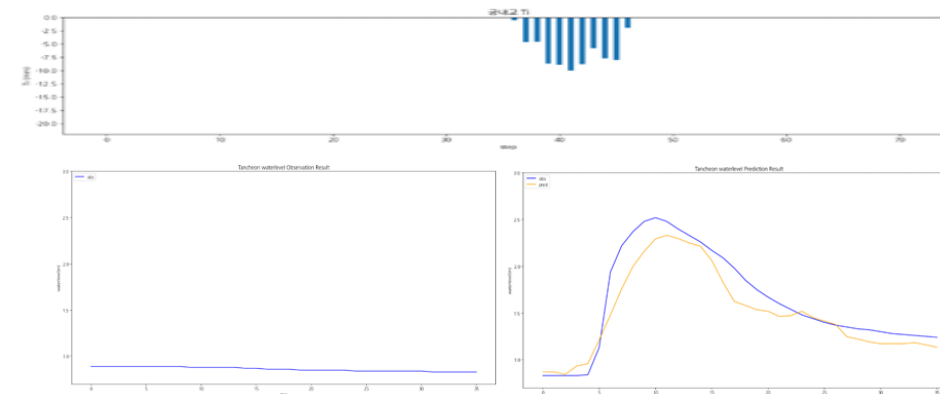
testdata_200802



testdata_220712



testdata_230711

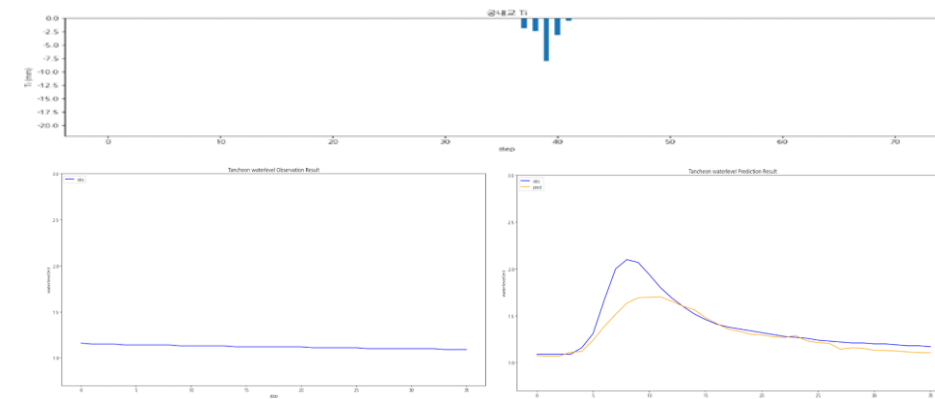


* model_(gn/dg)_huber

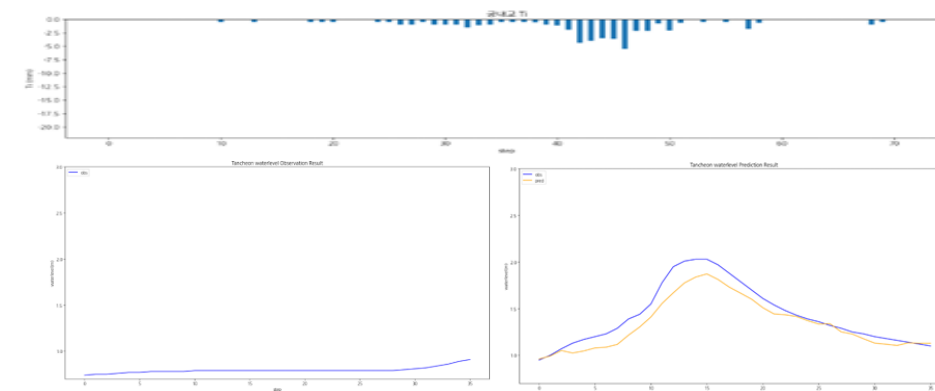
관심 수위 이상 + Huber + 0.0001
예측 코드 - **궁내교 2**

	궁내교
testdata_240723	r2 : 0.7258 mse : 0.0221 mae : 0.0855 rmse : 0.1489
testdata_250716	r2 : 0.8570 mse : 0.0143 mae : 0.0951 rmse : 0.1197
testdata_250814	r2 : 0.6947 mse : 0.0215 mae : 0.1217 rmse : 0.1467

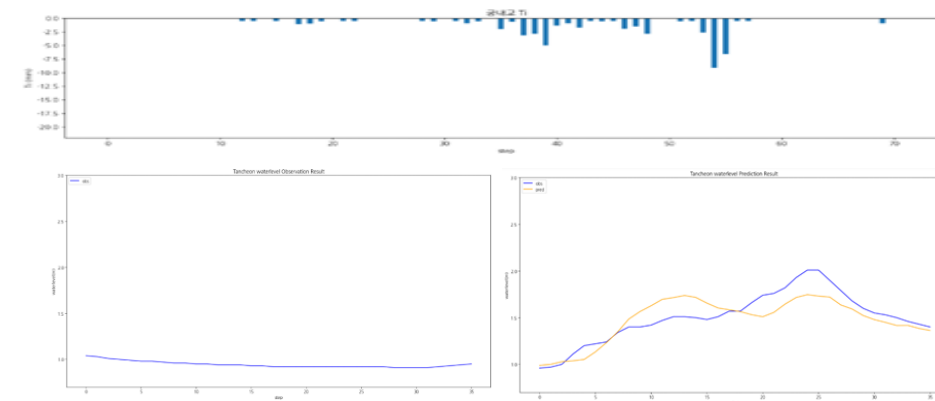
testdata_240723



testdata_250716



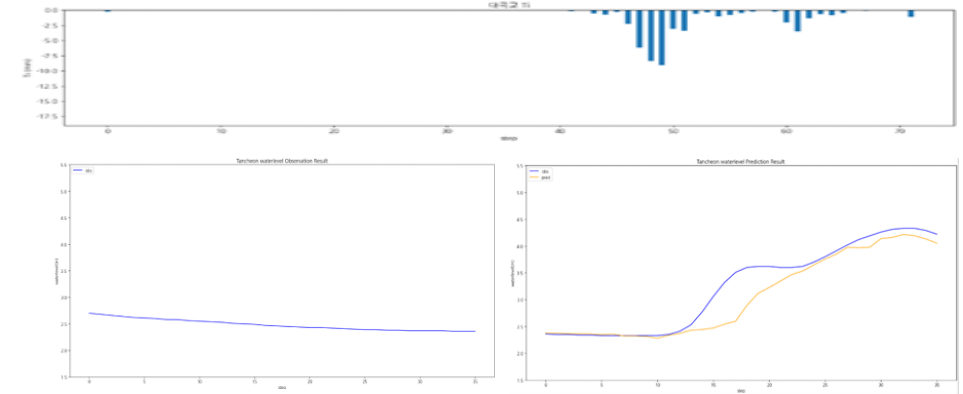
testdata_250814



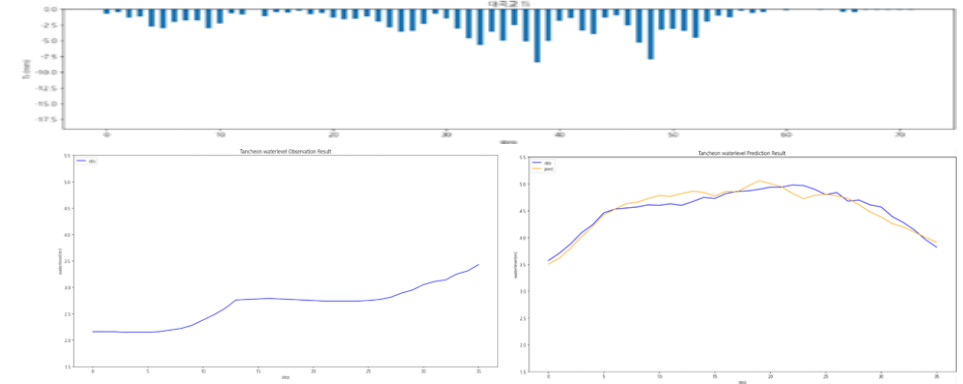
관심 수위 이상 + Huber + 0.0001 예측 코드 - 대곡교 1

	대곡교
testdata_200802	r2 : 0.8583 mse : 0.0864 mae : 0.1800 rmse : 0.2939
testdata_220712	r2 : 0.9114 mse : 0.0123 mae : 0.0931 rmse : 0.1113
testdata_230711	r2 : 0.8354 mse : 0.1497 mae : 0.2952 rmse : 0.3870

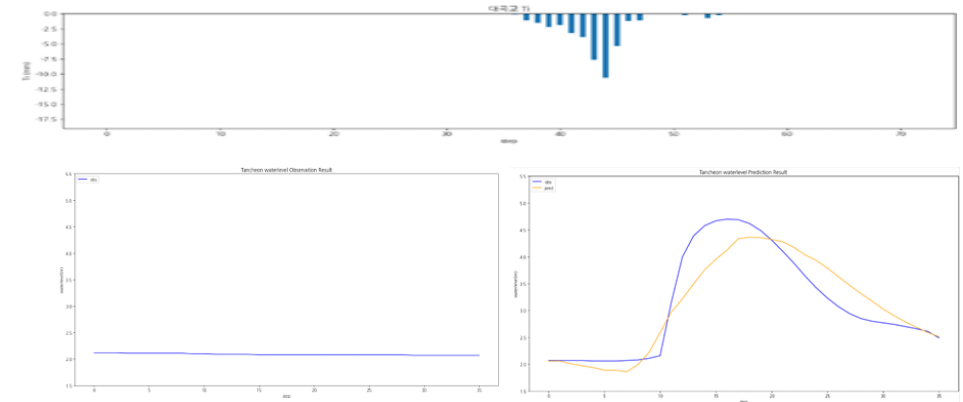
testdata_200802



testdata_220712



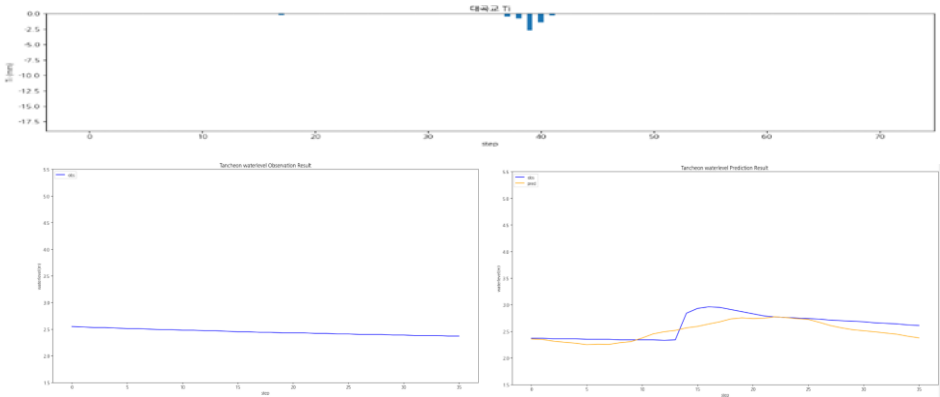
testdata_230711



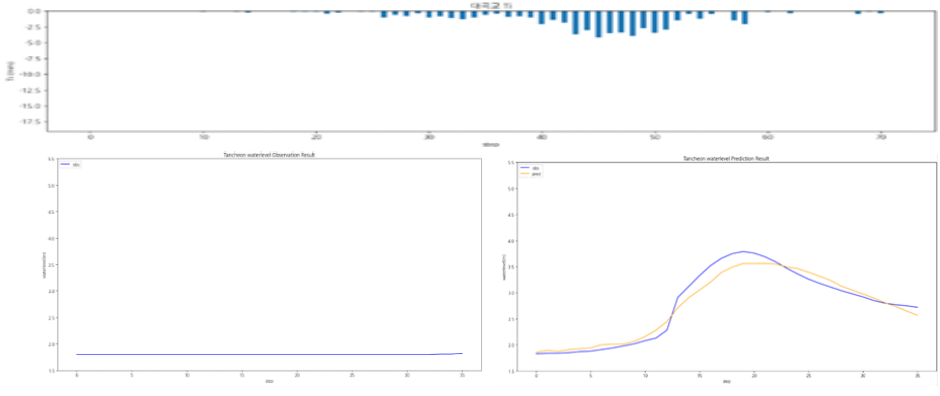
관심 수위 이상 + Huber + 0.0001
예측 코드 - 대곡교 2

	대곡교
testdata_240723	r2 : 0.5169 mse : 0.0229 mae : 0.1216 rmse : 0.1515
testdata_250716	r2 : 0.9566 mse : 0.0200 mae : 0.1153 rmse : 0.1414
testdata_250814	r2 : 0.7191 mse : 0.0974 mae : 0.2343 rmse : 0.3122

testdata_240723



testdata_250716



testdata_250814

